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Use implicit differentiation to find $\frac{dy}{dx}$.

$$2x^2y + 3xy^2 = 10$$

To use implicit differentiation, first differentiate both sides of the equation with respect to x , treating y as a differentiable function of x .

Use the rules of differentiation.

$$\frac{d}{dx}(2x^2y) + \frac{d}{dx}(3xy^2) = \frac{d}{dx}(10)$$

To find $\frac{d}{dx}(2x^2y)$, use implicit differentiation and the Derivative Product Rule.

$$\begin{aligned}\frac{d}{dx}(2x^2y) &= 2x^2 \frac{d}{dx}(y) + y \frac{d}{dx}(2x^2) \\ &= 2x^2 \frac{dy}{dx} + 4xy\end{aligned}$$

To find $\frac{d}{dx}(3xy^2)$, again use implicit differentiation and the Derivative Product Rule.

$$\begin{aligned}\frac{d}{dx}(3xy^2) &= 3x \frac{d}{dx}(y^2) + y^2 \frac{d}{dx}(3x) \\ &= 3x(2y) \frac{dy}{dx} + y^2(3) \\ &= 6xy \frac{dy}{dx} + 3y^2\end{aligned}$$

To find $\frac{d}{dx}(10)$, use the Derivative of a Constant Function Rule.

$$\frac{d}{dx}(10) = 0$$

Simplify.

$$\begin{aligned}\frac{d}{dx}(2x^2y) + \frac{d}{dx}(3xy^2) &= \frac{d}{dx}(10) \\ 2x^2 \frac{dy}{dx} + 4xy + 6xy \frac{dy}{dx} + 3y^2 &= 0\end{aligned}$$

Now, collect the terms with $\frac{dy}{dx}$ on one side of the equation.

$$2x^2 \frac{dy}{dx} + 6xy \frac{dy}{dx} = -4xy - 3y^2$$

Finally, factor out $\frac{dy}{dx}$ and then solve for $\frac{dy}{dx}$.

$$\begin{aligned}\frac{dy}{dx}(2x^2 + 6xy) &= -4xy - 3y^2 \\ \frac{dy}{dx} &= \frac{-4xy - 3y^2}{2x^2 + 6xy}\end{aligned}$$

Thus, using implicit differentiation, $\frac{dy}{dx} = \frac{-4xy - 3y^2}{2x^2 + 6xy}$.